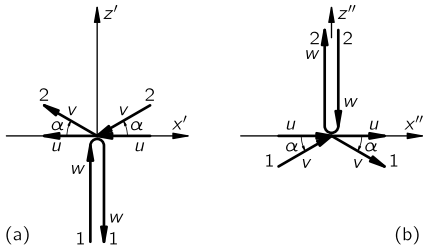

Figure 4-3 Two more views of the collision, from moving cars.

(2, because it moves up and back down). The obliquely moving particle has a certain velocity v whose components we have found to be u and $w\sqrt{1-u^2/c^2}$, and whose mass is m_v. The change in *vertical* momentum of this particle is therefore $\Delta p' = 2m_v w\sqrt{1-u^2/c^2}$ because, in accordance with our assumed law (4.8), the momentum component is always the mass corresponding to the magnitude of the velocity times the component of the velocity in the direction of interest. Thus in order for the total momentum to be zero the vertical momenta must cancel and the ratio of the mass moving with speed v and the mass moving with speed w must therefore be

$$\frac{m_w}{m_v} = \sqrt{1-u^2/c^2}\,. \tag{4.9}$$

Let us take the limiting case that w is infinitesimal. If w is very tiny indeed, it is clear that v and u are practically equal. In this case, $m_w \to m_0$ and $m_v \to m_u$. The grand result is

$$m_u = \frac{m_0}{\sqrt{1-u^2/c^2}}. \tag{4.10}$$

It is an interesting exercise now to check whether or not Eq. (4.9) is indeed true for arbitrary values of w, assuming that Eq. (4.10) is the right formula for the mass. Note that the velocity v needed in Eq. (4.9) can be calculated from the right-angle triangle:

$$v^2 = u^2 + w^2\,(1 - u^2/c^2).$$